

SRIVARDHAN BABA VEMULA

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Professional Summary

Cloud & Data Engineer with **3+ years** delivering cloud-native ETL pipelines, streaming systems, and AI-integrated applications on AWS and Azure. Achieved **40% throughput gains**, **65% faster batch runtimes**, and **99.8% uptime**; contributing to deepfake detection research under **Prof. Siwei Lyu** (IEEE Fellow, Director — UB Institute for AI & Data Science).

Technical Skills

Languages	Python, SQL, Java, R, Bash, Unix Shell Scripting
ML & AI	Deepfake Detection, Regression, Classification, scikit-learn, XGBoost, TensorFlow, Feature Engineering
Data Engineering	ETL/ELT Pipelines, Data Modeling, Schema Design, Data Quality, Batch & Stream Processing
Cloud, MLOps & Big Data	AWS (S3, EC2, Lambda), Azure (ADF, Azure SQL), PySpark, Kafka, Spark Streaming, Docker
Databases & Warehouses	PostgreSQL, MySQL, SQL Server, Snowflake, BigQuery
DevOps	CI/CD, Git, Docker, Sterling File Gateway
Orchestration & BI	Apache Airflow (DAGs, SLA monitoring), Tableau, Power BI, Excel Power Query

Education

University at Buffalo — M.S. Data Science

Aug 2025 – Dec 2026

Coursework: Machine Learning, Statistical Learning, Data Mining, Predictive Analytics, Probability, Database Systems, SQL, Numerical Computing

Experience

Graduate Research Assistant — AI & Data Systems

Jan 2026 – Present

University at Buffalo | UB Media Forensic Lab | Advisor: Prof. Siwei Lyu, IEEE Fellow, AI & DS Dept. Director

- Architected **Python ETL pipelines** loading **50K+ multi-modal records** into an **AWS S3 data lake**; slashing preprocessing errors by **45%** for **Deep-O-Meter** (zinc.cse.buffalo.edu).
- Engineered **PySpark distributed workflows** for feature extraction on **100GB+ datasets**; halved computation time by **60%** via adaptive partitioning and broadcast joins.
- Designed **ML feature pipelines** (PCA, SMOTE, normalization) for **5+ deepfake models**; lifting F1-score by **12%** through structured training data.
- Built **Flask REST APIs** for production ML inference and model serving; reduced latency from **580ms to 378ms** and improved throughput by **28%**.
- Established **MLflow model versioning, experiment tracking, and model lifecycle management** across **10+ configurations**; reduced QA effort by **30%**.

Cloud & Data Engineering Associate

Sep 2024 – Jul 2025

Broadridge Financial Solutions, India

- Automated **AWS S3 + SFTP ingestion workflows** processing **200+ files daily**; boosted throughput by **40%** and eliminated **15 hrs/week** of manual reconciliation.
- Integrated **Sterling File Gateway** with **5+ enterprise systems**; enforced SOX-compliant data lineage, lowering cross-system errors by **35%**.
- Constructed **SLA observability frameworks** (CloudWatch + PagerDuty) for **12+ pipelines**, sustaining **99.8% uptime**; trimmed incident resolution by **25%** via Airflow retry logic.
- Refactored **20+ SQL queries** to CTEs and window functions; restructured **4 schemas**, shrinking query latency by **30%** and storage overhead by **20%**.
- Instituted **CI/CD test suites** across **15+ workflows**, cutting post-deployment defects by **40%**.

Data Analyst & Python Developer

Aug 2022 – Aug 2024

Freelance, Remote

- Productionized **Python ingestion pipelines** processing **500K+ records** from **10+ sources**; reduced execution time by **45%** via vectorized transforms.
- Restructured relational schemas with composite indexing, achieving **60% faster queries** across **8 datasets** for 3 enterprise clients.
- Shipped **15+ Tableau and Power BI dashboards** saving clients **6+ hrs/week** in manual reporting.
- Maintained data workflows with validation and exception handling, sustaining **99.5% accuracy** over **2M+ records**.

Projects

- Real-Time Streaming Platform** (Kafka, Spark Streaming, AWS S3): Achieved **sub-2s latency** on **500K+ events/hour**; sustained **99.9% uptime** at 3× traffic spikes.
- Distributed Data Processing** (PySpark, Hadoop): Processed **100GB+ datasets**, halving batch runtime by **65%** via broadcast joins and adaptive partitioning.
- Azure Data Warehouse** (ADF, Snowflake, SQL): Unified **6 sources** into Snowflake transforming **50M+ records/month**; shortened load SLA from **4 hrs to 45 min**.
- Customer Analytics & Churn Prediction** (scikit-learn, XGBoost): Trained classifiers achieving **85% accuracy** via feature engineering and cross-validation.
- Workflow Orchestration Platform** (Airflow, PostgreSQL): Authored **10+ DAGs** eliminating **20+ hrs/week** manual scheduling; reached **98% SLA compliance**.

Certifications

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| • IBM Data Science Professional Certificate | • IBM Data Analyst Professional Certificate |
| • IBM Data Analysis & Visualization Foundations | • AWS Essentials for Developers |
| • Career Essentials in Microsoft Azure Fundamentals (AZ-900) | • Career Essentials in Microsoft Azure AI Fundamentals (AI-900) |
| • Google Crash Course on Python | • Infosys Young Python Professional |
| • Infosys Object-Oriented Programming Using Python | • Career Essentials in Generative AI — Microsoft & LinkedIn |